# Homozygous Familial Hypercholesterolemia (FH)

Homozygous FH is a condition resulting from defects in both alleles of the LDL receptor gene, leading to either complete absence or severely reduced LDL receptor activity.

## Clinical Findings: Cutaneous Lesions

Patients with homozygous FH typically exhibit severe xanthomatosis. Multiple types of xanthomata are present, including:

- **Tuberous xanthomata**, especially over the elbows
- **Subperiosteal xanthomata**, located below the knees and over the olecranon process
- **Tendon xanthomata**, commonly found in the Achilles tendon and extensor tendons of the hands
- **Elevated xanthomatous plaques**, distributed over the extremities, buttocks, and hands

Intertriginous xanthomas are rare but may occur in this condition. The rate of xanthoma formation correlates with the duration and severity of elevated LDL cholesterol levels.

## Laboratory Tests

In homozygous FH, serum LDL-cholesterol levels are markedly elevated, often ranging from 800 to 1000 mg/dL.

## Prognosis and Clinical Course

These patients frequently develop coronary heart disease (CHD) or other forms of atherosclerotic cardiovascular disease before adolescence.

### Treatment

Due to minimal or absent LDL receptor function, therapies that rely on upregulating LDL receptor activity—such as bile acid sequestrants and HMG-CoA reductase inhibitors (statins)—are generally ineffective. However, high-dose statins may produce modest reductions in LDL-cholesterol, likely through partial suppression of VLDL secretion.

More aggressive treatment strategies include:

- **Liver transplantation**, which provides functional LDL receptors from the donor organ
- **LDL apheresis** or **plasmapheresis**, procedures that periodically remove LDL from the circulation

These interventions are often necessary to achieve meaningful reductions in LDL levels and slow disease progression.

# Hypercholesterolemia (Frederickson Type II Hyperlipidemia)

### *At a Glance*

- Causes include **homozygous familial hypercholesterolemia**, **heterozygous familial hypercholesterolemia**, or **defective apolipoprotein B-100**

- Characterized by **severe hypercholesterolemia**, which may manifest clinically with **tendinous or tuberous xanthomas**
- Typically managed with **HMG-CoA reductase inhibitors (statins)**
- Associated with a **high risk of coronary atherosclerosis**

# Familial Heterozygous Hypercholesterolemia

## Epidemiology

Heterozygous FH is the more common form of the disorder, with a prevalence of approximately **1 in 500 individuals**.

## Etiology and Pathogenesis

It results from inheritance of a single mutated allele of the LDL receptor gene, leading to reduced LDL receptor function.

## Clinical Findings: Cutaneous Lesions

Unlike homozygous FH, tuberous xanthomata are uncommon in heterozygotes. However, **tendon xanthomata** are frequently observed, particularly in the Achilles tendon and extensor tendons of the hands. In fact, the majority of patients presenting with tendon xanthomata have heterozygous FH.

## Laboratory Tests

Heterozygous hypercholesterolemia is often detected during routine adult screening, with serum LDL-cholesterol levels approximately **twice the normal value**.

# Prognosis and Clinical Course

- **Men** with heterozygous FH typically develop coronary heart disease in their **30s to 40s**
- **Women** usually manifest CHD in their **50s to 60s**